

Vatsal Trivedi

Principal Engineer · AI Platforms · Enterprise Integrations

Bengaluru, India trivedivatsal005@gmail.com +91 760-063-5053 <https://vatsal.xyz>
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Profile

I'm a principal engineer who builds AI products and the platforms they run on. At Phamax, I lead Ariya's **architecture and delivery** across **eight pharma and life-sciences client deployments**. The platform handles source-backed generation, regulated content workflows, and enterprise integrations. Before that, I led engineering for Factoreal's omnichannel marketing product and Comviva's MobilYtix Rewards, working on analytics at scale, loyalty targeting, and event-driven services. I write code, set engineering standards, and lead teams through production releases. I also build open-source developer tools.

Focus areas Production RAG · LLM orchestration · AI evaluation · Tool calling & agent workflows · Regulated content lifecycle · Enterprise integrations · Platform engineering · Multi-tenant SaaS · Distributed systems · Event-driven architecture · Large-scale analytics · Data engineering · Observability & reliability · Application security · Developer tooling · Engineering leadership · Architecture & delivery

Experience

Phamax

Bengaluru, India
May 2025 - Present

Principal Engineer (Ariya Platform)

Ariya: AI platform for pharmaceutical content, research, analytics, and engagement

- Restructured Ariya into core API, portal, and authoring applications. Set **shared engineering standards** and reusable staging, UAT, and production deployment workflows.
- Built the AI Content Wizard and **regulated content lifecycle**, with configuration-driven authoring, localization, **multi-stage approval, compliance checks, and distribution** across content types and markets.
- Designed **Content Compose** with three LLM stages, hybrid Azure AI Search retrieval, **source-linked bibliographies** rebuilt after generation, and a **14-check validator** for completeness and citations.
- Designed **LLM orchestration** with prompts, context injection, RAG, embeddings, and tool calling. Integrated **Veeva CRM and Salesforce Marketing Cloud** into omnichannel workflows for each tenant.
- Delivered a white-label RAG assistant over **551 medical publications** with access controls, inspectable source PDFs, feedback, audit logs, and an admin console on Azure Container Apps.
- Built the **Speech Avatar** across Azure Speech, Django, and the portal, including streaming chat, credential exchange, conversation logs, and post-call analysis.
- Designed **Bronze/Silver/Gold ingestion** with a DBML registry and run tracking. Fixed six causes of production failures and added deterministic keys for **repeatable UPSERTs**.
- Built the **Data Catalog** for metadata discovery, observability, and **self-service access**, and the HCP and SFMC email-data APIs for the Daily Sales Brief.
- Built repeatable SonarQube checks and applied **20 security fixes** across **three client release lines** for encryption, JWT validation, hashing, SQL access, and application controls.
- Encrypted phone numbers at rest**, automated dependency security audits, and coordinated database certificate rotation across **seven environments and four services**.
- Added uv and Docker caching to Python builds. Built a **JSON-to-PPTX service** and admin APIs for content metrics and event reports.

Comviva

Bengaluru, India
October 2023 - March 2025

Technical Lead

MobilYtix Rewards: multi-tenant loyalty and campaign platform for telecom and retail

- Led the **cloud-agnostic, multi-tenant platform rewrite** in TypeScript, React, and FeathersJS, with PostgreSQL, Delta Lake, and vector databases. Built product features and the services and data systems they use.
- Owned **Kafka campaign processing** for high-throughput reward and loyalty workflows, with asynchronous execution and caching. Delivered **AI/ML personalization and segmentation**.
- Built a **JSON-to-PostgreSQL segmentation engine** with 26 operators, including 11 date operators. Added **custom member attributes without schema changes**, scheduled runs, and run history.
- Made **club provisioning transactional** so partial tenant setup rolls back. Used Kafka to recompute segments when their rules change.
- Wrote PostgreSQL and Redis **clients used by five services** and service-to-service JWT helpers. Built a shared member package for enrollment, membership validation, attribute updates, and caching.

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- Introduced **platform observability and monitoring** for more reliable deployments and faster incident diagnosis. Coordinated releases and security fixes across six services.
- Introduced Jest tests and coverage gates for segmentation, scheduling, and tiers, including **48 segmentation tests**. Moved shared packages to TypeScript and automated publishing.

Factoreal

Bengaluru, India

Technical Lead

December 2021 - September 2023

Factoreal: omnichannel marketing automation platform, incubated by Comviva

- Built and expanded BigQuery analytics for a platform handling **10 billion transactions**. Automated provisioning and scoring across **71 customer datasets and 998 scheduled queries**.
- Designed the **Fan Maturity Model** for the sports vertical. Six channel scores and customer-configurable weights produce four audience tiers, with daily scoring and portal controls.
- Rebuilt the **Shopify app for its App Store relaunch**: installation, signup, store provisioning, and OAuth. Expanded webhook coverage from **17 to 40 topics** and added GraphQL billing subscriptions.
- Built **SeatGeek ticketing synchronization** and WhatsApp commerce data capture. Kept Shopify API compatibility current and fixed duplicate contacts with **country-aware phone normalization**.
- Built shared **contact operations across five services** and **plan entitlements across twelve**. Automated customer retirement and improved contact queries and data-cleanup routines.
- Added **tenant-configurable session expiry and risk-scored signup protection**. Automated GoodData cache invalidation and built social dashboards and Jira-to-BigQuery reporting sync.

Factoreal

Bengaluru, India

Senior Engineer

June 2020 - November 2021

Factoreal: omnichannel marketing automation platform, incubated by Comviva

- Built **PostgreSQL and Cassandra-to-BigQuery synchronization** across 24 entity workflows, with **incremental processing and pagination**. Automated customer dataset provisioning at signup.
- Built **email delivery and engagement KPI APIs** and embedded GoodData dashboards so customers can see deliverability, audience response, and unsubscribe rates.
- Made Shopify order ingestion safe under retries with **application idempotency and database uniqueness constraints**. Built **Redis- and Kafka-driven abandoned-cart triggers** with tenant-configurable timeouts.
- Published **Razorpay payment events into Kafka**. Built web-push subscriptions and mobile SDK registration and events for commerce and engagement workflows.
- Fixed **tracking accuracy** with page deduplication, tenant-specific event exclusions, and first-visit corrections. Built verified-domain trial onboarding and **HubSpot contact migration**.

Tata Consultancy Services (TCS)

Gandhinagar, India

Assistant System Engineer

September 2018 - May 2020

Life sciences work for global clients.

- Built **React and Angular applications** with Node.js and Express backends for enterprise life-sciences workflows.
- Built **Microsoft 365 workflows** with SharePoint, Teams, Power Automate, and Power BI for collaboration, process automation, and reporting.
- Set up **GA360, Google Tag Manager, and Data Studio analytics** so life-sciences teams can measure digital activity and performance.

Projects I built

Personal projects and developer tools I built, including open-source software.

openpreflight

openpreflight.xyz

Self-hosted GitHub Check Runs for private pull requests. One Go binary, one SQLite file, on a server you run. Apache-2.0.

Go · SQLite · Docker · GitHub Apps

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Knock

knock.codes

Copy-paste React and Tailwind components for private previews: access screens, invitations, and draft controls. Drops into a shadcn project.

React · Tailwind · shadcn/ui

InboxCraft

inboxcraft.app

A **browser-only PowerShell generator** for Microsoft 365 Outlook inbox rules. Quick and Advanced modes, with templates for GitHub, Slack, and Jira. Scripts stay in the browser.

TypeScript · React · PowerShell

InboxCraft Skills

github.com/trivedi-vatsal/inboxcraft-skills

Official Agent Skills pack for InboxCraft. **Nine PowerShell tools** so assistants can manage Outlook rules, folders, and categories from a prompt.

PowerShell · Agent Skills · Outlook

audit-plan-tasks

github.com/trivedi-vatsal/audit-plan-tasks

Agent Skill that scaffolds AUDIT.md, PLAN.md, and TASKS.md so a plan is based on what you actually checked, not on the request.

Agent Skills · Planning

StaleGuard

vatsal.xyz/StaleGuard

Reloads a stale SPA when the tab is focused again, by checking ETag and Last-Modified. No polling and no banner.

TypeScript · React

PySecRecipes

vatsal.xyz/pysec-recipes

GitHub Actions recipes that **audit Python dependencies** against PyPI, OSV, and Python Packaging advisories, and open or close issues as findings change.

Python · GitHub Actions · DevSecOps

PyImportSync

github.com/trivedi-vatsal/PyImportSync

AST check that **every Python import is declared** in requirements.txt. Ships as a GitHub Action and a pre-commit hook.

Python · AST · pre-commit

Skills

Core Areas - AI Platforms · Enterprise Integrations · Multi-Tenant SaaS · LLM Systems · Distributed Systems · Event-Driven Architecture · Developer Tooling · Data Engineering · Application Security · Regulated Environments · Platform Architecture · Engineering Leadership · Regulated Content Workflows

AI & LLM Engineering - LLM Orchestration · RAG Pipelines · Prompt Engineering · Tool Calling · AI Evaluations · Azure OpenAI · Azure AI Search · OpenAI API · Claude · LangChain · LangGraph · CopilotKit / AG-UI · Azure Speech · Vector Databases · MCP · Agent Skills

Platform Engineering - CI/CD · GitHub Actions · GitLab CI · Docker · Azure (App Service, Functions, Blob Storage, Key Vault) · Bicep · SonarQube · Observability · Infrastructure as Code · Security Automation · Azure Container Apps

Programming Languages - TypeScript · JavaScript · Python · Go · SQL · PowerShell

Frameworks & Tools - React · Next.js · Node.js · Django · FastAPI · Celery · Strapi · FeathersJS · Express · Keycloak · Kafka · BullMQ · Jest · GraphQL & REST · Vercel AI SDK · Shopify · Turborepo · shadcn/ui

Databases - PostgreSQL · MySQL · SQLite · MongoDB · BigQuery · Redis · Cassandra · Delta Lake

Data Engineering - Medallion Architecture · ETL & Sync Pipelines · Pandas · PyArrow · SQLAlchemy · GoodData · Data Catalogs · Metadata Discovery

Design - Figma · Affinity Designer · Procreate

Enterprise Integrations - Veeva CRM · Salesforce Marketing Cloud · Shopify · SeatGeek · HubSpot · Razorpay

Education

Bachelor of Engineering, Computer Engineering 2014 - 2018

Birla Vishvakarma Mahavidyalaya Engineering College, Vallabh Vidyanagar